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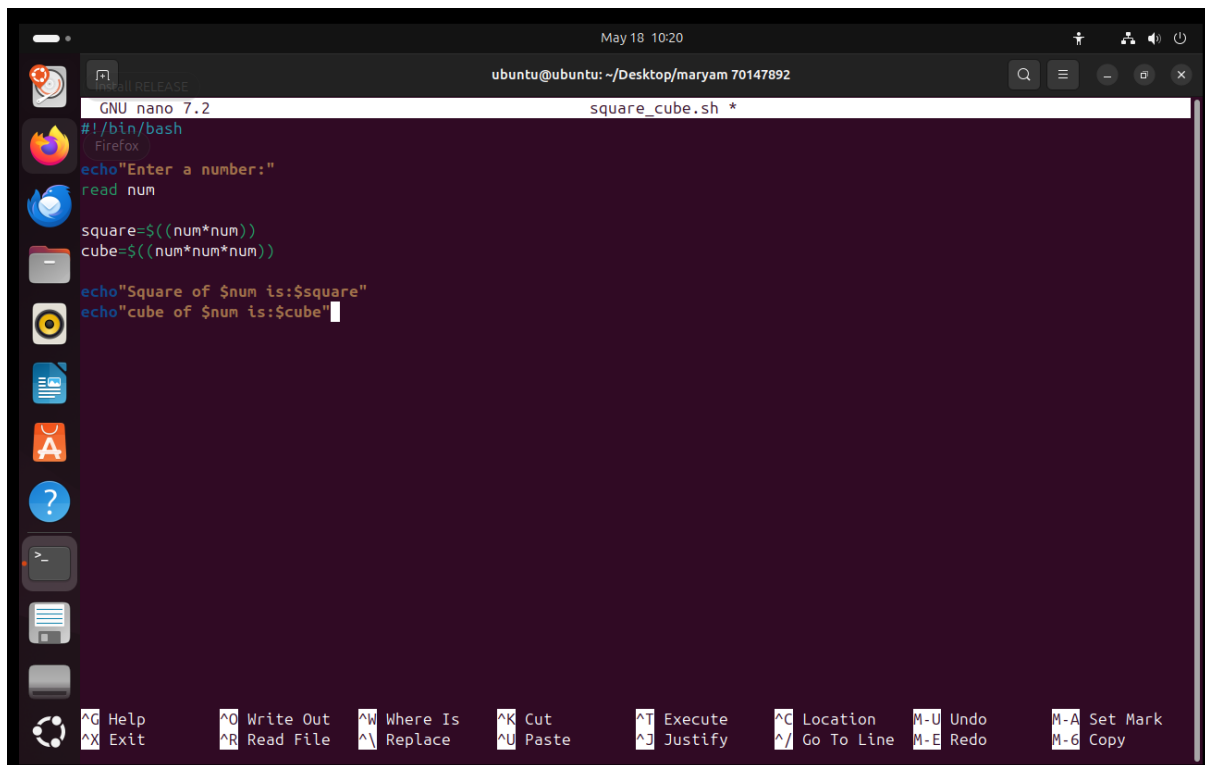
Lab Task

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Section: S

Subject: Operating System

Example 1:



The screenshot shows a terminal window with the nano text editor open. The editor is editing a file named `square_cube.sh`. The script content is as follows:

```
#!/bin/bash
echo "Enter a number:"
read num

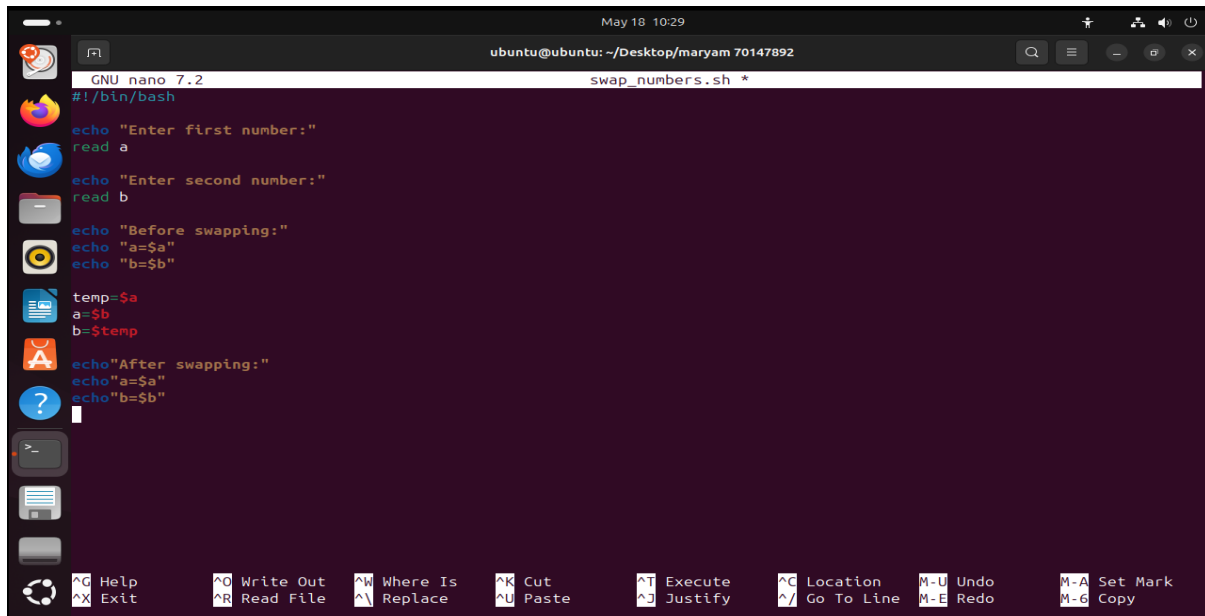
square=$((num*num))
cube=$((num*num*num))

echo "Square of $num is:$square"
echo "cube of $num is:$cube"
```

The terminal window title bar shows the user is `ubuntu@ubuntu` and the current directory is `~/Desktop/maryam 70147892`. The nano editor status bar at the bottom shows various keyboard shortcuts.

```
ubuntu@ubuntu:~/Desktop/maryam 70147892$ nano square_cube.sh
ubuntu@ubuntu:~/Desktop/maryam 70147892$ chmod +x square_cube.sh
ubuntu@ubuntu:~/Desktop/maryam 70147892$ ./square_cube.sh
Enter a number:
8
Square of 8 is:64
cube of 8 is:512
ubuntu@ubuntu:~/Desktop/maryam 70147892$
```

Example 2:



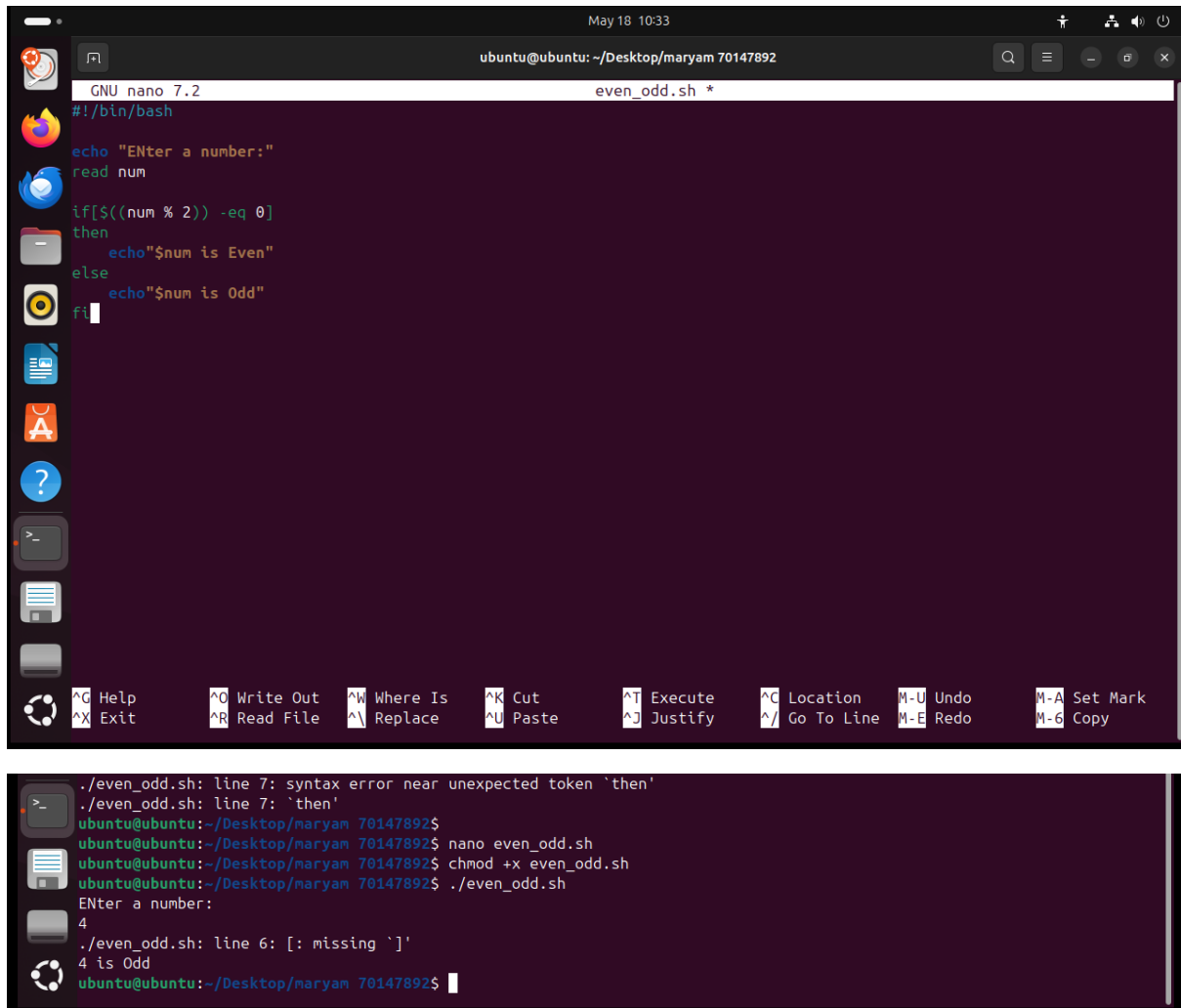
The screenshot shows a terminal window with the nano text editor open. The editor is editing a file named `swap_numbers.sh`. The script contains the following code:

```
#!/bin/bash
echo "Enter first number:"
read a
echo "Enter second number:"
read b
echo "Before swapping:"
echo "a=$a"
echo "b=$b"
temp=$a
a=$b
b=$temp
echo "After swapping:"
echo "a=$a"
echo "b=$b"
```

The nano editor's status bar at the bottom shows various keyboard shortcuts: Help Exit, Write Out Read File, Where Is Replace, Cut Paste, Execute Justify, Location Go To Line, Undo Redo, and Set Mark Copy.

```
ubuntu@ubuntu:~/Desktop/maryam 70147892$ nano swap_numbers.sh
ubuntu@ubuntu:~/Desktop/maryam 70147892$ chmod +x swap_numbers.sh
ubuntu@ubuntu:~/Desktop/maryam 70147892$ ./swap_numbers.sh
Enter first number:
8
Enter second number:
3
Before swapping:
a=8
b=3
./swap_numbers.sh: line 17: echoAfter swapping: command not found
./swap_numbers.sh: line 18: echoa=3: command not found
./swap_numbers.sh: line 19: echob=8: command not found
ubuntu@ubuntu:~/Desktop/maryam 70147892$
```

Example 3:

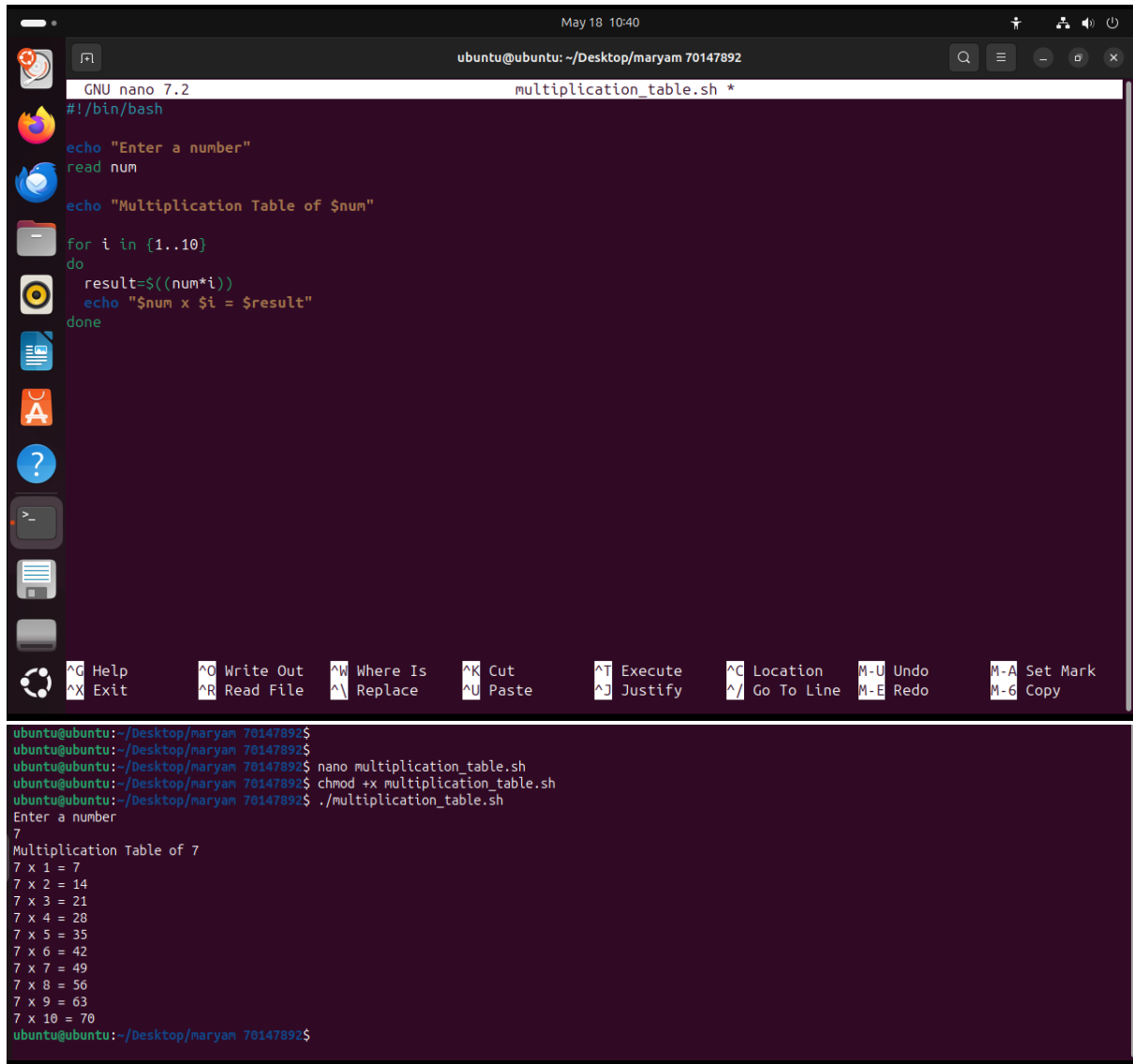


```
May 18 10:33
ubuntu@ubuntu: ~/Desktop/maryam 70147892
GNU nano 7.2 even_odd.sh *
#!/bin/bash
echo "ENter a number:"
read num
if[ $((num \% 2)) -eq 0$ ]
then
    echo"$num is Even"
else
    echo"$num is Odd"
fi

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo     M-A Set Mark
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-G Copy
```

```
./even_odd.sh: line 7: syntax error near unexpected token `then'
./even_odd.sh: line 7: `then'
ubuntu@ubuntu:~/Desktop/maryam 70147892$ nano even_odd.sh
ubuntu@ubuntu:~/Desktop/maryam 70147892$ chmod +x even_odd.sh
ubuntu@ubuntu:~/Desktop/maryam 70147892$ ./even_odd.sh
ENter a number:
4
./even_odd.sh: line 6: [: missing `]'
4 is Odd
ubuntu@ubuntu:~/Desktop/maryam 70147892$
```

Example 4:



The screenshot shows a terminal window with a dark background and a light-colored terminal font. The window title is "ubuntu@ubuntu: ~/Desktop/maryam 70147892". The terminal displays the following content:

```
GNU nano 7.2 multiplication_table.sh *
#!/bin/bash
echo "Enter a number"
read num
echo "Multiplication Table of $num"
for i in {1..10}
do
    result=$((num*i))
    echo "$num x $i = $result"
done
```

The terminal also shows the execution of the script:

```
ubuntu@ubuntu:~/Desktop/maryam 70147892$
ubuntu@ubuntu:~/Desktop/maryam 70147892$ nano multiplication_table.sh
ubuntu@ubuntu:~/Desktop/maryam 70147892$ chmod +x multiplication_table.sh
ubuntu@ubuntu:~/Desktop/maryam 70147892$ ./multiplication_table.sh
Enter a number
7
Multiplication Table of 7
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70
ubuntu@ubuntu:~/Desktop/maryam 70147892$
```

The terminal window includes a sidebar with various application icons and a bottom status bar with keyboard shortcuts for nano editor commands.